TO BE CHOSEN

Data Science 1130: Final Project Report

BY: Kaosisisochukwu Nweke , Mohammad Alkharoubi , Amalo Adusei

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Overall Research Topic: Which consumer and loan attributes impact the likelihood of default?

Background/introduction of problem:

In this credit EDA analysis we used data sets from 'applications_data', 'columns_description', and 'previous_applications' to determine how we can identify loan applicants who may have difficulty making payments, perhaps leading to measures such as refusing loans, lowering loan amounts, or offering higher interest rates to risky applicants? The ultimate goal was to ensure that individuals capable of repaying their loans are not denied, with data-driven insights used to identify these specific candidates. We choose to dive deep into two specific areas which are the impact of Income-to-Loan Amount Ratio (DTI) on repayment difficulties and how the different Education_type to Repayment Difficulties.

Data Cleaning

Before diving into the data we imported all necessary libraries and created code to check for missing values in an automated manner. It started by counting and calculating the percent of missing values in each column of the data frames application_df and the previous_df . We created a column 'missing_data_columns' that stores columns with a missing value percentage of 40% or greater. Using the 'drop()' method, the missing columns were detected and then removed from the combined data frame. We then produced a new dataframe 'combined_data' which was cleaned and combined the two data frames . Finally, it uses an assertion check to ensure that no rows in the new dataframe have more than 50% missing values. In addition, since one of the hypotheses requires us to explore applicants with a higher total income, a decision was made to analyse those who are in the top 5 percentile. The salary chosen as the minimum for high was $162,610 it was chosen according to spring financial.ca. On top of that, when calculating the DTI to avoid division by 0 any 0s in income was replaced with 1 to avoid division by zero

<u>Subtopic 1: The Impact of Income-to-Loan Amount Ratio (DTI) on Repayment Difficulties</u>

Introduction

Our analysis embarked on testing the hypothesis that "Applicants with a higher total income but a high ratio of loan amount to income are more likely to experience repayment difficulties compared to those with a lower income-to-loan amount ratio.” This hypothesis suggests that within the segment of top earners, those with disproportionately high debt relative to their income may be more likely to face financial issues, potentially leading to repayment difficulties which are the people we want to avoid giving out loans to.

Data Analysis

The data investigation involved focusing on the group which is the top 5% earners, focusing on their debt-to-income (DTI) ratios and their relationship with loan repayment outcomes, captured by the 'TARGET' variable in our dataset. Through visual analytics, particularly boxplots and violin plots, we were able to identify a clear trend which is that individuals facing repayment difficulties had noticeably higher median DTI ratios than their counterparts. This finding was a direct reflection of our hypothesis, suggesting a link between higher DTI ratios and repayment challenges.

The density plots revealed that high debt-to-income (DTI) ratios don't always lead to repayment issues. It was observed that individuals with higher DTI percentages tend to struggle more with repayments, aligning with initial expectations. On the other hand, the scatter plot showed otherwise as for a group of high earners despite their high DTI ratios, they were managing their repayments well. This could imply that other factors, such as having access to savings or additional income sources, may affect the risk associated with a high DTI.

Conclusion

In sum, our analysis lends support to the hypothesis by demonstrating that a higher loan-to-income ratio among the top 5% earners is indeed associated with a greater likelihood of repayment difficulties. However, the relation between DTI ratios and repayment outcomes is not linear and moderated by other outside factors. The relationship is not strictly linear, as suggested by the variations in repayment difficulties among those with similar DTI ratios. Additionally, our hypothesis testing indicated that the relationship between DTI ratio and repayment difficulties is not accurate due to the p-value of zero which we got as our result. Therefore, while the DTI ratio is a critical measure for assessing repayment risk, other factors should be considered for a more comprehensive financial structure for an accurate predictive analysis.

Drawbacks of Our Analysis:

- Our study suggests that the debt-to-income (DTI) ratio is an important sign of how likely someone is to have trouble paying back loans. But this view might be too simple because it doesn't consider all the different bills and costs people have, which can also impact their ability to make loan payments.
- Assumptions on High Earners' Financial Behavior: We assume that high earners' financial behaviours are consistent with the general population, which may not be the case.
- Outlier Influence on Findings: High-income outliers or those with extremely high loan amounts may influence our analysis, potentially affecting the perceived relationship between income, loan amount, and repayment difficulty.

<u>Sub-Topic 2: Educational level influence on loan repayment</u>

Introduction

There may be a link between a client's educational level and their ability to repay. Clients having more education, such as those with academic degrees, may demonstrate better repayment behaviour than those with less education. This hypothesis also looks into whether clients with more education prefer to apply for larger loan amounts than those with less education.

Data Analysis

We wanted to discover the relationship between educational levels of a client and repayment issues among loans by comparing education and loan payback. Our countplot research looked into whether particular education categories had higher percentages of repayment difficulties. A box plot analysis was also performed to see whether higher-educated persons tend to apply for larger loan amounts than those with lower education levels. These findings are intended to assist financial institutions in tailoring lending strategies depending on clients' educational backgrounds and financial capacities.

We discovered that educated people have more difficulty repaying their loans. However, the increased percentage of educated borrowers made drawing definitive conclusions purely from the numbers difficult. To solve this, we combined datasets, filtered for employed persons, and used a line graph to illustrate job counts across education groups. Notably, those with a higher education worked fewer days on average and median than those with a lower education, indicating a substantial difference in maximum and lowest days worked between these groups. We did hypothesis testing and a correlation study between education levels and loan repayment to corroborate these findings.The maximum and minimum days worked vary significantly among these education groups. We conducted a hypothesis test to be sure that the comparison is significant and that we don’t judge just based on numbers. Then we conducted a correction test to find out if there was a correlation between education level and loan repayment.

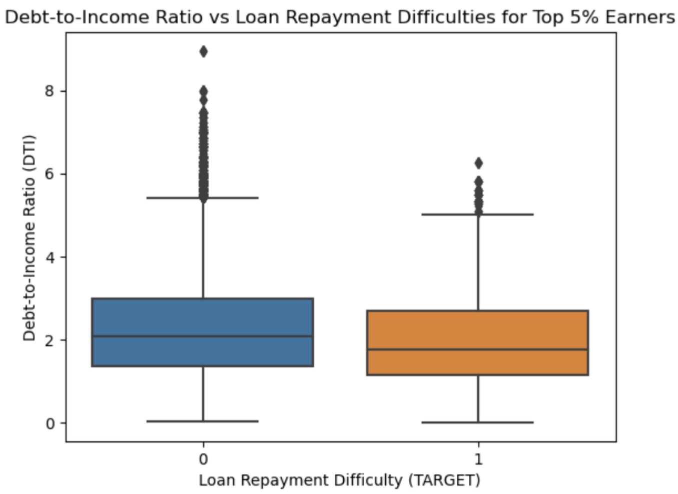
Conclusion

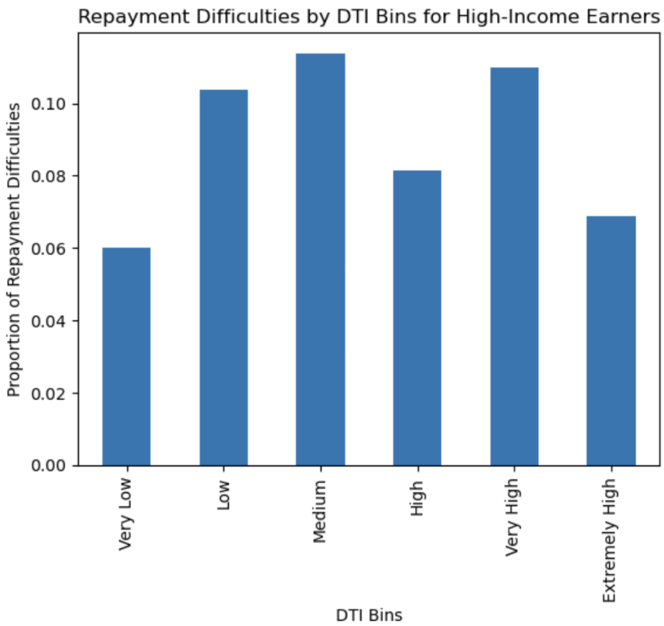
To conclude we performed a linear regression and correlation test after cleaning and analysing the NAME_EDUCATION_TYPE and TARGET. The study reveals a possible relationship between education level and loan repayment behaviour, with higher educational achievement associated with better payback performance. Furthermore, it may imply that clients with higher levels of education are more reliable in repaying loans, which could indirectly imply that they are applying for greater loan amounts, but this specific portion would require a direct analysis of loan amount data by education level to validate.

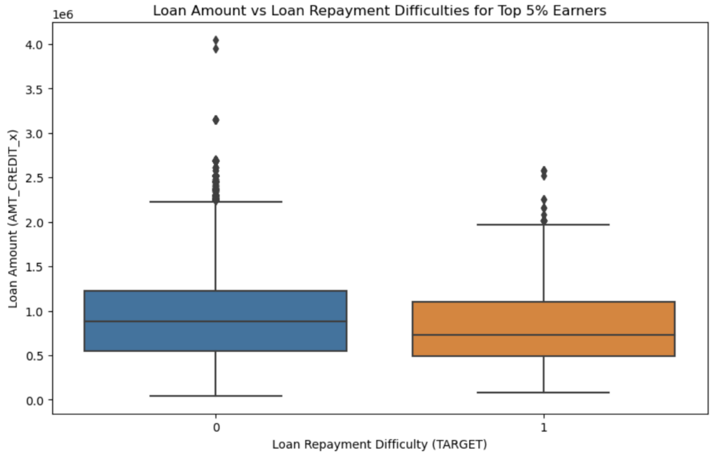
Drawbacks of Our Analysis:

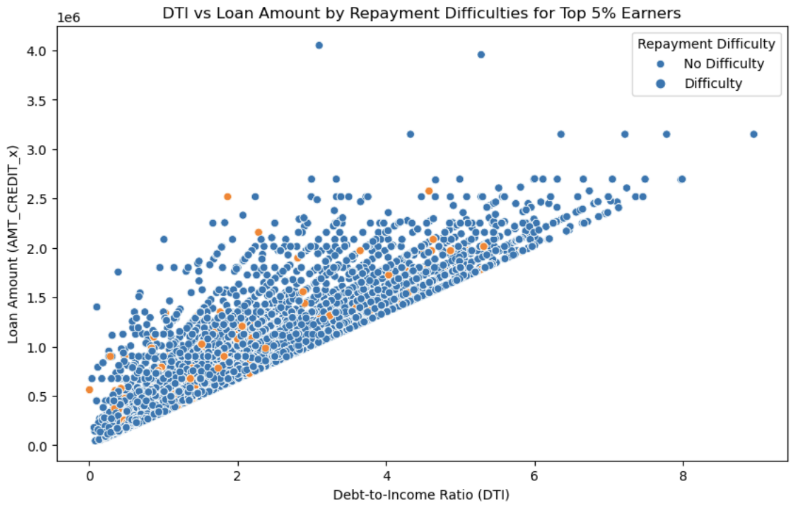
According to my research, the client's educational degree influences loan payback. However, this observation lacks thorough specifics because there may be other lesser-known education-related characteristics that contribute to a person's difficulties repaying a loan that are likely neglected for instance, employment status which was one of the characteristics we evaluated.

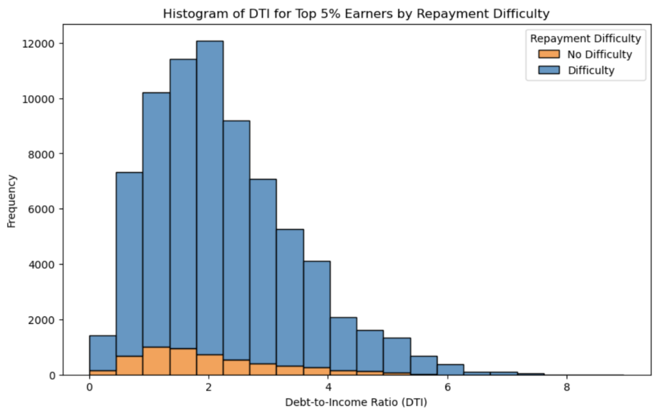
Sub-Topic 1 visualisation

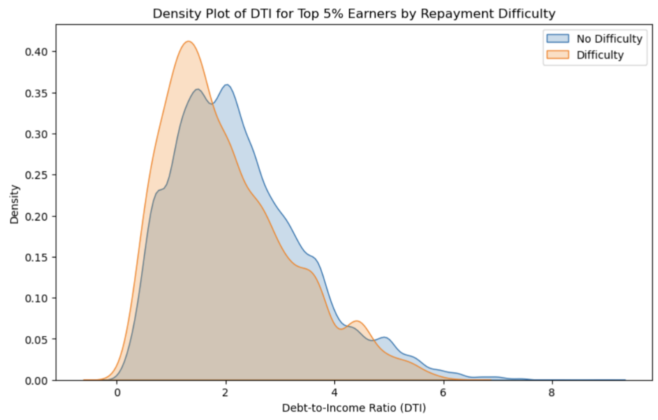


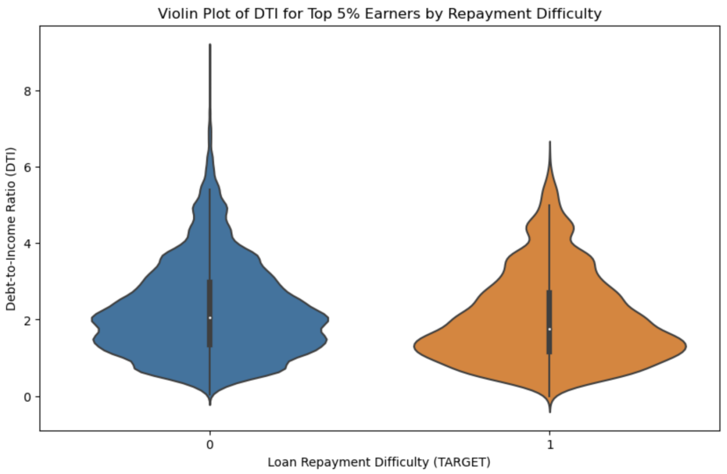



Loan Amount vs Loan Repayment Difficulties for Top 5% Earners
1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Loan Amount (AMT_CREDIT_x)
0
1
Loan Repayment Difficulty (TARGET)

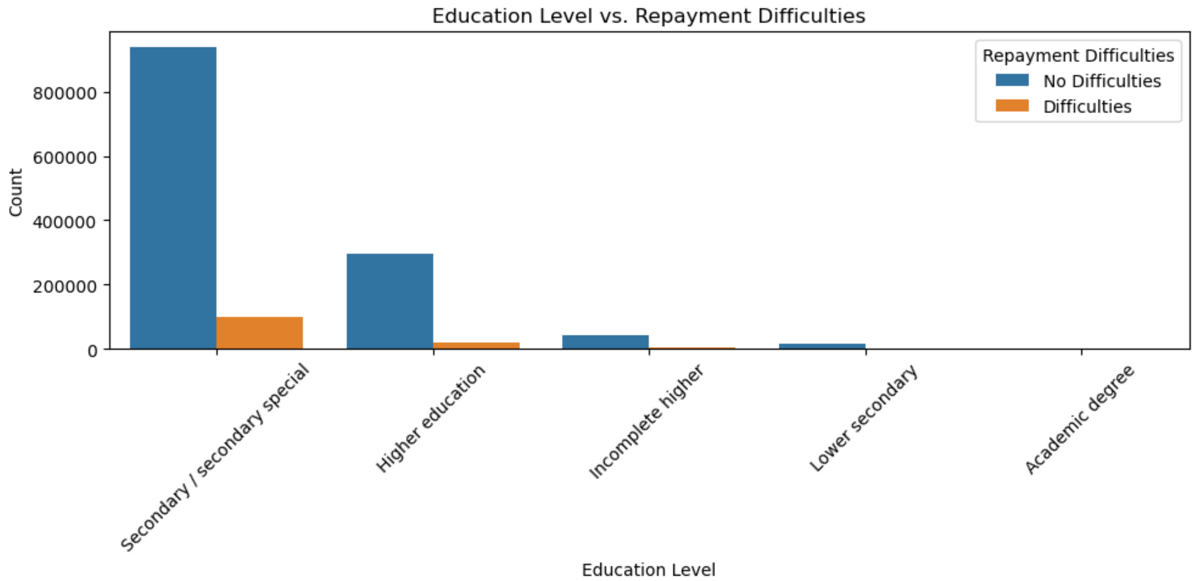

DTI vs Loan Amount by Repayment Difficulties for Top 5% Earners
1e6
Repayment Difficulty
No Difficulty
Difficulty
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Loan Amount (AMT_CREDIT_x)
0
2
4
6
8
Debt-to-Income Ratio (DTI)

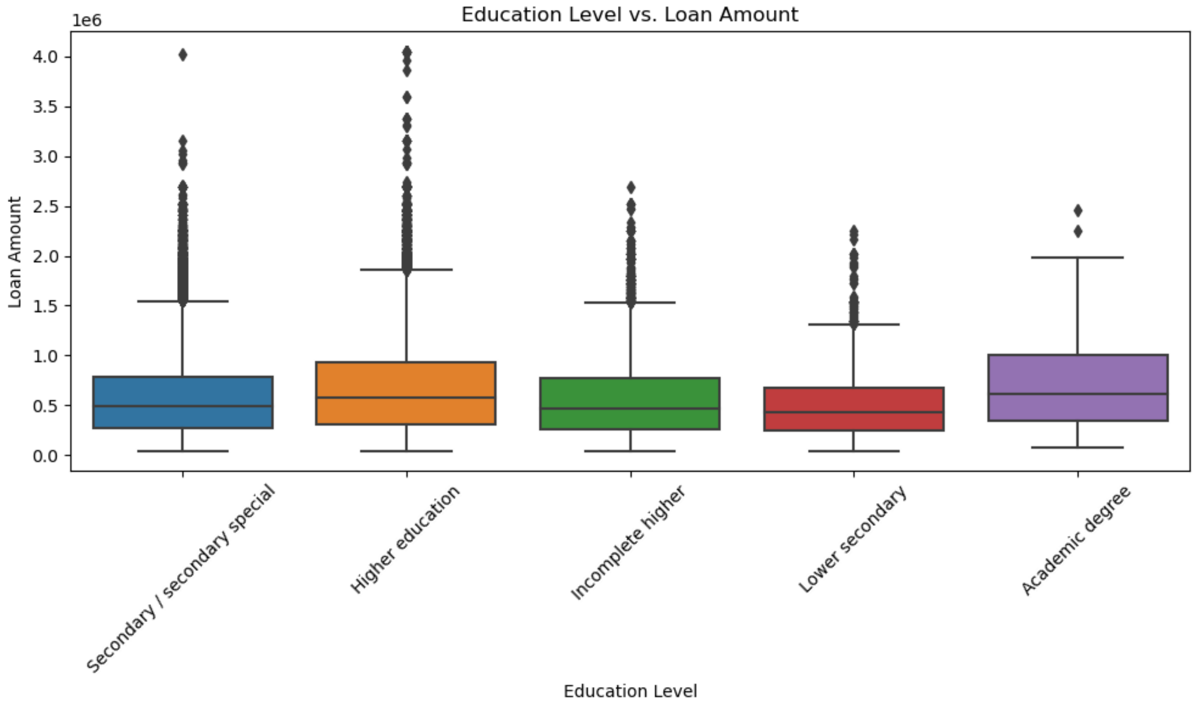

Histogram of DTI for Top 5% Earners by Repayment Difficulty
Repayment Difficulty
No Difficulty
Difficulty
12000
10000
8000
6000
4000
2000
0
Frequency
0
2
4
6
8
Debt-to-Income Ratio (DTI)

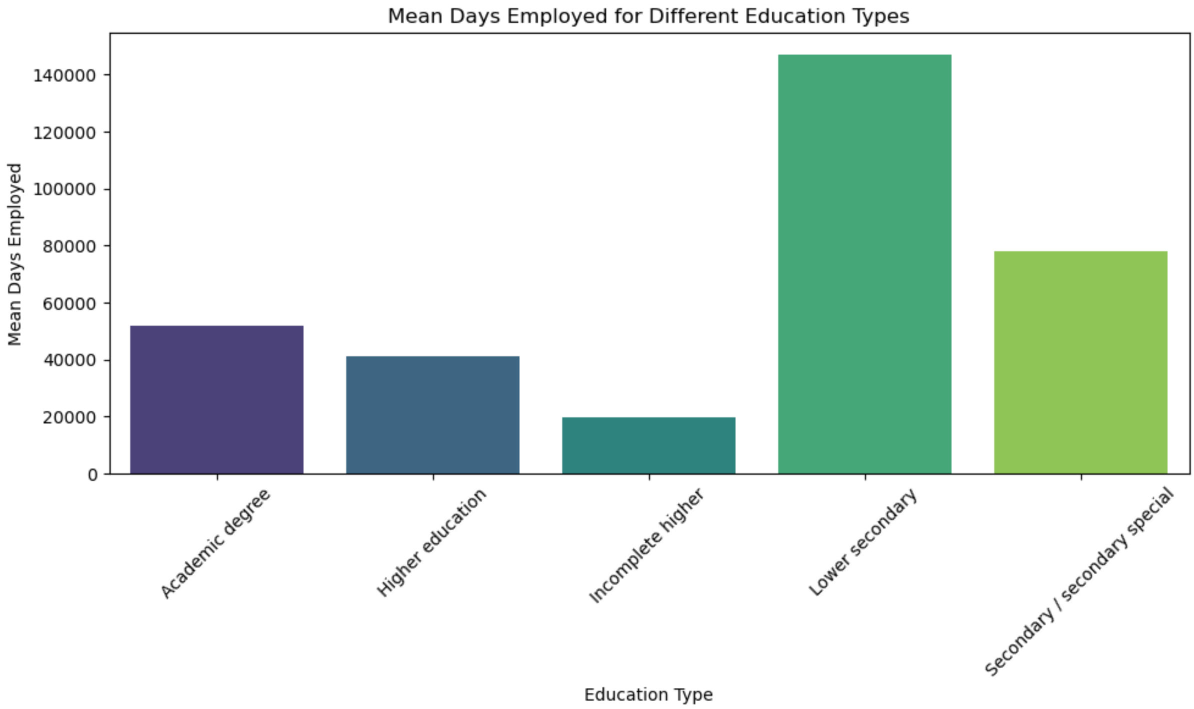


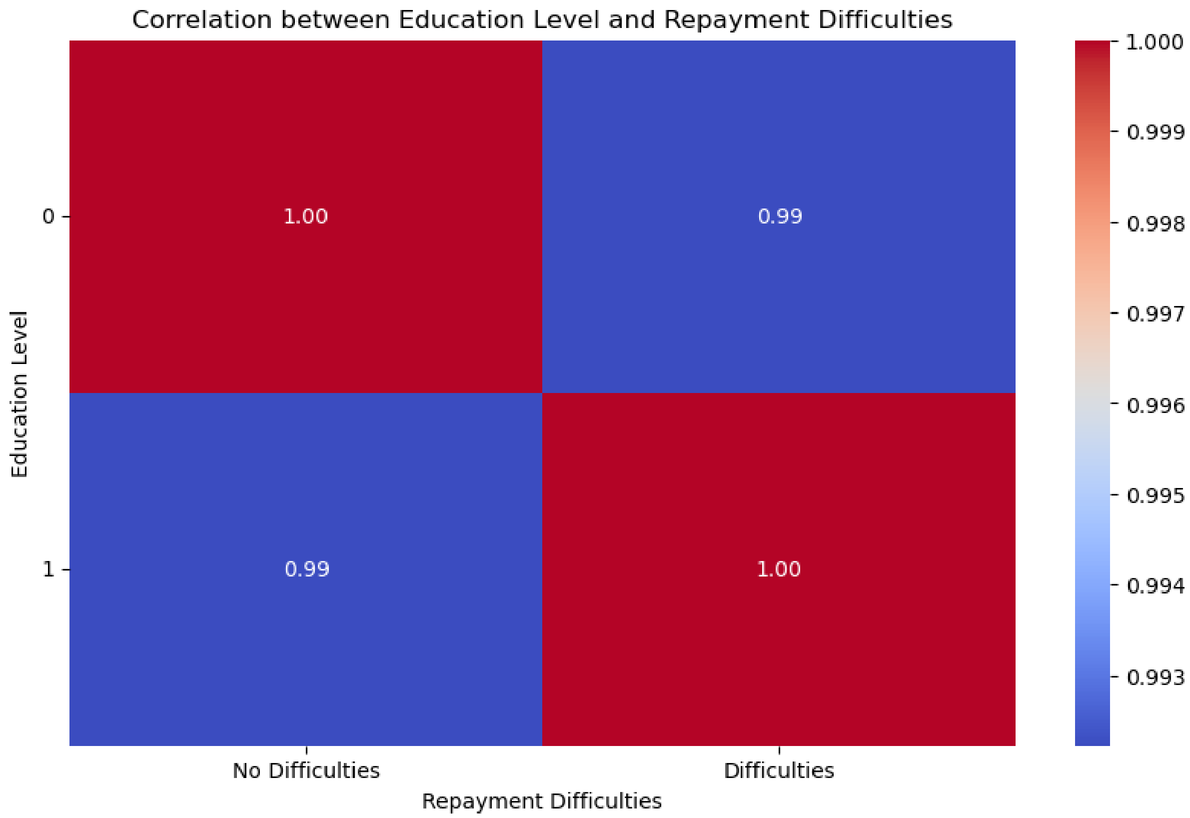


Sub-Topic 2 visualisation


Education Level vs. Repayment Difficulties
Repayment Difficulties
No Difficulties
Difficulties
Count
800000
600000
400000
200000
0
Secondary / secondary special
Higher education
Incomplete higher
Lower secondary
Academic degree
Education Level


Education Level vs. Loan Amount
1e6
Loan Amount
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Secondary / secondary special
Higher education
Incomplete higher
Lower secondary
Academic degree
Education Level


Mean Days Employed for Different Education Types
Mean Days Employed
140000
120000
100000
80000
60000
40000
20000
0
Academic degree
Higher education
Incomplete higher
Lower secondary
Secondary / secondary special
Education Type


Correlation between Education Level and Repayment Difficulties
Education Level
0
1
1.00
0.99
0.99
1.00
No Difficulties
Difficulties
Repayment Difficulties
1.000
0.999
0.998
0.997
0.996
0.995
0.994
0.993